ECON 0100 | HW C3 | Corrective Policies

Solutions

Answers and work are shown in the red blocks.

Due: Sunday, Oct. 18

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | Malodorous Byproducts

Butterbeer is beloved by Hogwarts students and teachers alike and is widely available at wizarding shops. But the beverage has the negative effect of generating displeasing odors from its imbibers. Preferences for butterbeer and its marginal cost can be represented by the following relationships.

$$D : P_B = 100 - \frac{1}{2}Q_D \qquad S : P_S = 20 + \frac{1}{2}Q_S$$

Prices are in galleons and quantity is in bottles. The odor imposes a cost of 20 galleons per bottle on everyone nearby. Using your economic policy advising credentials, use a graph and some algebra to offer a remedy to the malodorous byproduct that the Ministry could implement.

Policy (tax or subsidy): _____ Size per bottle: _____

Market Quantity ($\hat{Q}$): _____ Quantity with Your Policy: _____

Policy: a per-unit (corrective, or Pigouvian) tax. Size: 20 galleons per bottle. Market quantity: 80 bottles. Quantity with your policy: 60 bottles.

Market. Buyers and sellers ignore the odor, so the market settles where demand meets supply:

$$100 - \tfrac{1}{2}Q = 20 + \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 80, \quad P = 60$$

Efficient quantity. Each bottle costs the brewer $20 + \frac{1}{2}Q$ and costs everyone nearby 20 more, so the marginal social cost is $\text{MSC} = 40 + \frac{1}{2}Q$. Society wants bottles only while their marginal benefit covers that:

$$100 - \tfrac{1}{2}Q = 40 + \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 60$$

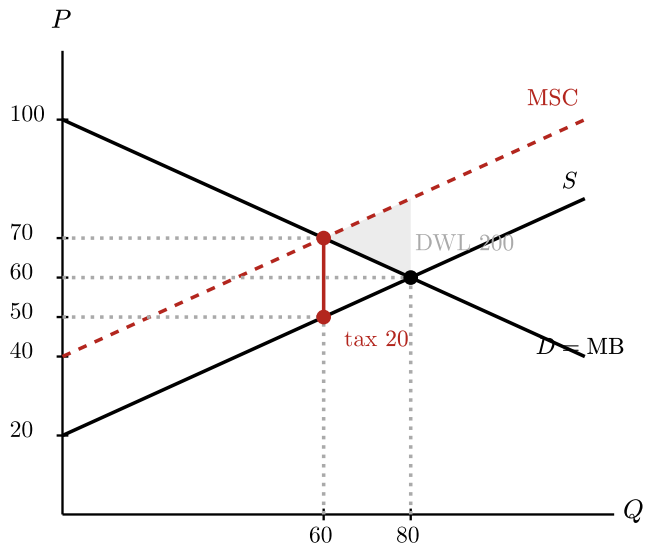


Figure 1: Corrective tax

Why a tax of 20. The market overproduces because sellers leave 20 of each bottle's cost out of their decision. A tax of exactly 20 per bottle hands that cost back to them: to sell a bottle they now need its private cost *plus* 20, which is the marginal social cost. The market then stops where marginal benefit equals marginal social cost, the same quantity a social planner would choose. The tax wedge of 20 only fits between demand and supply at $Q = 60$.

Effects on the market (for reference; not asked). Buyers pay $100 - 30 = 70$, up 10; sellers keep $20 + 30 = 50$, down 10; the Ministry collects $20 \times 60 = 1200$. Bottles 61 through 80 cost society more than they are worth to buyers, so the market quantity carried a deadweight loss, the gray triangle between MSC and demand:

$$\text{DWL} = \tfrac{1}{2} \times 20 \times (80 - 60) = 200$$

The tax eliminates it. The odor still exists, but only on bottles worth their full cost.

Q2 | Dragon Pox

Dragon Pox is a virus that has spread widely throughout the wizarding world. Gorsemoor's Dose prevents a witch or wizard from catching it, meaning they also don't pass Dragon Pox on to anyone else. The marginal private benefit of a dose is $P = 24 - Q$, and supply, which is both the private

and the social cost, is $P = 4 + Q$. Prices are in galleons and quantity is in doses. Each dose also gives a benefit of 6 galleons to everyone else, since they are less likely to catch Dragon Pox.

Offer a remedy the Ministry could implement to correct the market failure. And conclude with a description of the effects your policy would have on the market for Gorsemoor's Dose.

Policy (tax or subsidy): _____ Size per dose: _____

Market Quantity ($\hat{Q}$): _____ Quantity with Your Policy: _____

Policy: a per-unit (corrective, or Pigouvian) subsidy. Size: 6 galleons per dose. Market quantity: 10 doses. Quantity with your policy: 13 doses.

Market. Buyers count only their own benefit, so the market settles where marginal private benefit meets supply:

$$24 - Q = 4 + Q \quad \Rightarrow \quad Q = 10, \quad P = 14$$

Efficient quantity. Each dose is worth $24 - Q$ to the drinker and 6 more to everyone else, so the marginal social benefit is $\text{MSB} = 30 - Q$. Society wants doses as long as that covers the cost:

$$30 - Q = 4 + Q \quad \Rightarrow \quad Q = 13$$

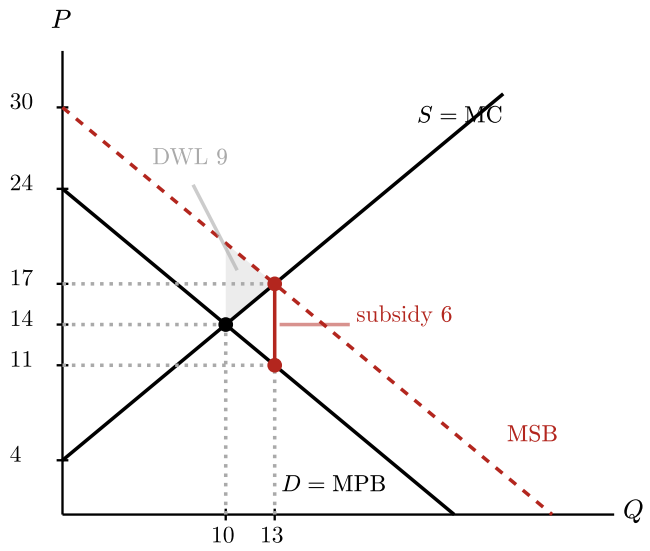


Figure 2: Corrective subsidy

Why a subsidy of 6. The market underproduces because buyers leave the 6 of benefit to everyone else out of their decision. A subsidy of exactly 6 per dose pays them that benefit: a dose is now worth its private benefit *plus* 6 to the buyer, which is the marginal social benefit. The market then stops where marginal social benefit equals marginal cost, the same quantity a social planner would choose. The subsidy wedge of 6 only fits between supply and demand at $Q = 13$.

Effects on the market (for reference; not asked). Buyers pay $24 - 13 = 11$, down 3; sellers receive $4 + 13 = 17$, up 3; the Ministry spends $6 \times 13 = 78$. Doses 11 through 13 are worth more to society than they cost, so the market quantity carried a deadweight loss, the gray triangle between MSB and supply:

$$\text{DWL} = \tfrac{1}{2} \times 6 \times (13 - 10) = 9$$

The subsidy eliminates it.

How does your policy bring the market to the socially efficient quantity?

a) It lowers the cost of brewing each dose, so sellers offer fewer.

b) It raises the price buyers pay, so fewer witches and wizards drink it.

c) Buyers now receive the benefit to everyone else on every dose, so their benefit plus the subsidy matches marginal social benefit, and the market settles where marginal social benefit equals marginal cost.

d) It sets a price floor, so sellers earn more on every dose.

c)

The subsidy adds the 6 of benefit to everyone else to what each dose is worth to its buyer, so demand plus the subsidy is MSB and the market lands where MSB = MC. a) A lower cost would mean *more* doses, not fewer. b) The subsidy lowers the price buyers pay, from 14 to 11, so more of them drink it. d) is a price floor, which is not the policy.